

A Case of Upper Eyelid Defect with Human Bite Successfully Managed with Cutler-Beard Procedure

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ABSTRACT

A 30-year-old woman presented to our hospital emergency with a right upper eyelid injury caused by a human bite, sustained 6 hours prior. Examination revealed a full-thickness defect, affecting 70% of the upper eyelid. The patient was successfully treated using the Cutler-Beard procedure, resulting in satisfactory functional and cosmetic outcomes.

Keywords: Cutler-Beard flap, lid sharing surgery, Human bite, upper eyelid defects

INTRODUCTION

The Cutler-Beard procedure, also known as the bridge flap technique, was first introduced in 1955 by Norman Cutler and Crowell Beard to reconstruct significant upper eyelid defects where more than 50% of the lid margin is involved, and the Tenzel semicircular flap procedure is insufficient.¹ This surgical approach is commonly employed to repair large upper eyelid defects caused by congenital conditions (such as colobomas), trauma, or complications following tumor excision.¹⁻² The primary goal of the surgery is to restore the normal anatomical and functional integrity of the upper eyelid. The procedure is a two-stage advancement lid-sharing surgery, typically performed under local or general anesthesia depending on the patient's age, health status, and the surgeon's expertise. Unfavorable outcomes reported with this technique include lower lid ectropion due to cicatrization, absence of upper eyelid eyelashes, and upper eyelid entropion due to lack of tarsal support.³⁻⁵ Preventative measures for these complications include using tarsal substitutes like nasal septum cartilage, ear cartilage,³ autogenous tarsal plate, preserved sclera, or conjunctiva from the opposite lid, or employing a rotation flap to avoid entropion.

The absence of eyelashes after the procedure can be addressed with artificial eyelash implantation or a lash flap technique.⁶ However, the Cutler-Beard procedure is not recommended for adolescents with monocular vision as it requires the eye to be closed for 6–8 weeks.

It also not recommended for young children below 7 years of age with monocular or binocular vision because of its potential to cause amblyopia.⁶

The case report is done to create awareness and effectiveness in using Cutler-Beard procedure in the treatment of large upper eyelid defect (2/3 defect).

CASE REPORT

A 30-year-old woman presented to our hospital's emergency department with a 6-hour-old trauma to the right upper eyelid caused by a bite from another individual during a domestic altercation. She complained of bleeding from the right eye and pains but no associated loss of vision. Following the injury, her eye was rinsed with water by her neighbour. Unprescribed eye drops from a chemist were applied and the eye covered with cloth.

On examination, the patient appeared anxious but was conscious and alert. Visual acuity was 6/6 in both eyes. The right upper eyelid showed a full-thickness loss involving two-thirds of the eyelid, with a defect measuring 20 mm by 4 mm and mild lid edema. Other examinations of the right eye and the left eye were normal. The patient was admitted, and the right eye was cleaned with normal saline and povidone-iodine (figure 1).

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She was administered a tetanus toxoid injection (0.5 ml IM), IV Augmentin 500 mg every 12 hours for 48 hours, oral Metronidazole 400 mg three times daily for 5 days, Ibuprofen 200 mg three times daily for 3 days, and Chloramphenicol eye ointment at night for 2 weeks.

Routine investigations, including full blood count, urinalysis, electrolytes and urea, and retroviral screening, showed no abnormalities.

The patient underwent the Cutler-Beard procedure three days after admission, following informed consent, under general anesthesia as requested.

The Cutler-Beard procedure is a two-stage advancement lid-sharing surgery, typically performed under local or general anesthesia. The first stage: A width of the upper eyelid defect is measured either with ruler or caliper and the width is marked on the lower eyelid. A horizontal incision 2mm below the inferior tarsal plate is made with surgical blade and the incision extended using scissor to full thickness. The lower lid Conjunctival is dissected from lower eyelid retractor and brought under the bridge of lower lid and sutured to the edge of upper eyelid conjunctiva and levator aponeurosis. Second stage is usually performed 6-8 weeks after first procedure to separate the sharing lids using surgical blade and scissor. The horizontal lower lid incision is sutured to the cut edge of the lower eyelid.

Postoperatively (figure 2), she was prescribed oral Augmentin 500 mg every 12 hours for one week, vitamin C 200 mg three times daily for three weeks and continued her preoperative antibiotics apart for IV augmentin. She was discharged one week after the first stage of Cutler- Beard procedure.

Eight weeks later, the patient returned for the second stage of the procedure (figure 3). Her postoperative course was uneventful, and she achieved good functional and cosmetic results (figures 4 & 5).

DISCUSSION

Eyelids are crucial for maintaining the physiological function of the eye by protecting and preserving corneal integrity. Structural and functional defects in the upper eyelid can result in exposure keratopathy. Several techniques have been reported for reconstructing upper eyelid defects exceeding 50%.¹⁻⁶ However, the chosen method should minimize ocular tissue irritation and maintain corneal integrity. In our case, no complications were observed following the Cutler-Beard surgery. Nevertheless, previous reports have documented upper eyelid entropion and lower lid ectropion.²⁻⁴ The early intervention in our patient might have contributed to the absence of complications in this case. Long-term follow-up may be necessary to assess lid function and potential complications associated with this procedure. However, there was a loss of eyelashes on the right upper lid because the flap used in the procedure did not include eyelashes; this was also noted in previous studies.²⁻⁴

CONCLUSION

The Cutler-Beard procedure remains a valuable option for reconstructing large upper eyelid defects of up to 70%. The procedure is generally well-tolerated and provides satisfactory anatomical, functional, and cosmetic outcomes.



Figure 1: Patient at presentation showing right upper eyelid defect



Figure 2: First day of postoperative care (first stage Cutler- Beard procedure)



Figure 3: Second stage of the Cutler- Beard procedure at 8 weeks



Figure 4: One month after the second stage Cutler- Beard procedure operation



Figure 5: Patient at presentation showing right upper eyelid defect with Lagophthalmos



Figure 6: One month after second stage of Cutler- Beard procedure. No Lagophthalmos

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